Amon Saka Kaneko

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PROFESSIONAL SUMMARY

Computer Engineering student and Student Research Assistant focused on applied AI, computer vision, and edge deployment. Built YOLO26-based space situational awareness models, deployed an isotonic-calibrated XGBoost risk screener, and integrated sensor fusion with ROS 2 navigation.

TECHNICAL SKILLS

AI and machine learning: Python, computer vision, YOLO26, scikit-learn, XGBoost, LightGBM, LangChain, isotonic calibration, model quantization

Software development: C++, JavaScript/TypeScript, React, Flask, Git, Linux

Embedded and robotics: STM32, real-time operating systems (RTOS), Raspberry Pi CM5, Hailo, ROS 2, Nav2, 3D LiDAR, sensor fusion, Extended Kalman Filter

RF and hardware: Software-defined radio (SDR), SDR++, RF experimentation, telemetry, IoT devices, KiCad, 3D modelling software

PROFESSIONAL EXPERIENCE

Student Research Assistant (SSA Edge-AI Payload) | Sep 2026-Present

- Developing embedded software for an on-orbit space situational awareness payload using a low-cost commercial off-the-shelf (COTS) camera, Raspberry Pi Compute Module 5 (CM5), and Hailo accelerator.
- Cleaned and preprocessed image data; built, trained, and evaluated YOLO26-based object detection models alongside other candidate models for space situational awareness.
- Quantizing models for edge inference and evaluating detection performance against device constraints; programming redundant multi-boot and network-boot recovery paths for the CM5 payload.
- Exploring custom AI models for hyperspectral analysis as part of the payload research scope.

Advanced Robotics and Automation Centre (ARAC) - Student Assistant Engineer | Autonomous Navigation

- Benchmarked 2D and 3D LiDAR, RGB-D, infrared, and ultrasonic sensing on transparent surfaces; identified optical return dropout, narrow ultrasonic field of view, and off-axis specular reflection as coverage limits.
- Reimplemented the TOPGN pipeline: projected 3D LiDAR intensity into low, middle, and high 2D grid layers; isolated transparent-obstacle neighborhoods; extrapolated tangent-line obstacle boundaries; and accumulated detections across robot poses for mapping.
- Extended detection with ultrasonic ranging and an Extended Kalman Filter, combining LiDAR bearing with sonar range and retaining obstacle estimates during sonar dropout.
- Integrated transparent-obstacle detection into a ROS 2 Nav2 costmap layer for path planning and collision avoidance; completed 10 of 10 collision-free autonomous runs in a mock healthcare ward under varied lighting.

Artistry Heals - Software Team Lead

- Led an AI and immersive-technology platform for emotional awareness through creative arts from concept to MVP while supporting partnerships, customer discovery, pitching, and fundraising.
- Won Highest Social Impact Startup at Princeton TigerLaunch 2025; placed 2nd Runner-Up at NTU Port63; reached the KSIP 2025 Top 10 and incubated into the SMU BIG January 2025 cohort.

EDUCATION

Singapore Polytechnic - Diploma in Computer Engineering | Apr 2025-Present
Academic standing: GPA 4.00/4.00; SP Engineering Scholar; Director's Honour Roll AY25/26; Siemens Prize

Singapore Polytechnic - Polytechnic Foundation Programme, Science and Technology | Apr 2024-Feb 2025
Academic standing: GPA 4.00/4.00; Top PFP Student AY24/25; PFP Study Award; Director's Honour Roll AY24/25

SELECTED PROJECTS

Diabetes Risk Prediction and Clinic App | Python, scikit-learn, Flask

- Cleaned the CDC BRFSS 2015 survey from 441,456 raw responses to a 240,184-record modelling dataset; built leakage-safe preprocessing and stratified train, validation, and test splits.
- Tuned and compared nine models across eight families using cross-validation, ROC-AUC, and precision at matched recall; selected XGBoost at a recall-first operating point and applied isotonic probability calibration.
- Audited recall by race, sex, income, and age; used bootstrap confidence intervals to assess uncertainty. Packaged the calibrated model with threshold and feature-order metadata, then deployed a Flask risk questionnaire on Render.

Project: github.com/energelpen/ET0737-MLAI-Project | Questionnaire: mlai-project.onrender.com

BusTech Autonomous Vehicle Passenger Assistance System | Feb 2026-Present

- Co-developed a digital twin of an autonomous bus and a computer-vision passenger assistance system using existing onboard cameras; designed sensing logic to support passengers with different assistance needs.
- Combined edge computing with telemetry and IoT device inputs to provide onboard context for passenger assistance workflows.
- Implemented, integrated, and tested open language models within the passenger assistance system alongside its computer-vision and sensing components.
- Advanced to the BusTech grand finals.

RoboErectus and RoboCup Interest Group | CCA | Preparing for RoboCup 2027

- Automated image labelling for the computer vision training workflow; the trained model was deployed on the RoboErectus robot at RoboCup Korea.
- Designed and built a 6-DOF robotic arm from scratch, including its 3D model and a dedicated software package.
- Won third place at RoboRoarZ Indonesia as part of the RoboCup Interest Group team.

Fake News Detection Agent | Jan 2025 | LangChain

- Built a LangChain-orchestrated agent that analyses news content and produces a fake-news assessment through a structured multi-step workflow.

VOLUNTEERING AND COMMUNITY LEADERSHIP

SourceFoods - Founder

- Co-founded a community food-aid initiative that helps partner organisations track donated food and redistribute surplus before it is wasted; secured funding through grants and partner outreach.
- Led developers building a shared system for donations, inventory, and distribution, including tools that flag expiring stock and replace paper logs and disconnected spreadsheets.

Singapore Youth AI - Inspire Program Lead | Volunteer | Apr 2026-Present

- Lead school outreach and coordinate volunteers and school staff to deliver workshops on AI technology, career pathways, and responsible AI for secondary students.
- Develop reusable slide decks, lesson plans, and activities so facilitators can run consistent sessions across schools.

YouthTechSG - Acting Deputy Head of Programmes

- Grew online youth engagement 200% in six months through interactive events with external technology partners.
- Supported 20+ students through the YouthxSkills Cybersecurity Scholarship with SANS Institute and partner companies; helped place 50+ Open Impact fellows on live open-source projects.

ADDITIONAL

RF and licences: Licensed amateur radio operator with hands-on RF and SDR work using SDR++; IMDA Amateur Radio Station Licence (9V1KA); FCC General Class Licence (KF0RKQ)

Languages: English and Hindi (fluent); Japanese (proficient)